```
syms a b c
A = [1 1 1 1 1 1 ;
    2 3 1 1 1 1 ;
1 2 a 1 1 1;
2 3 a+1 b 2 2;
1 1 1 2 c 1;
3 4 a+2 b+1 c+1 3]
```

A =

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 2 & 3 & 1 & 1 & 1 & 1 \\ 1 & 2 & a & 1 & 1 & 1 \\ 2 & 3 & a+1 & b & 2 & 2 \\ 1 & 1 & 1 & 2 & c & 1 \\ 3 & 4 & a+2 & b+1 & c+1 & 3 \end{pmatrix}$$

```
IND = [8;
11;
10;
15;
12;
19]
```

IND = 6×1

```
     8
    11
    10
    15
    12
    19
```

```
AMP = [A IND]
```

AMP =

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 8 \\ 2 & 3 & 1 & 1 & 1 & 1 & 11 \\ 1 & 2 & a & 1 & 1 & 1 & 10 \\ 2 & 3 & a+1 & b & 2 & 2 & 15 \\ 1 & 1 & 1 & 2 & c & 1 & 12 \\ 3 & 4 & a+2 & b+1 & c+1 & 3 & 19 \end{pmatrix}$$

```
det(A)
```

ans = 0

```
rank(A)
```

```
ans =
5
```

```
rank(AMP)
```

```
ans =
6
```

La matriz es de 6*6 y tiene rango 5 y la amp 6, por tanto es incompatible

Pero una submatriz de 5*5 si que puede ser compatible

```
B = A(1:5,1:5)
```

B =

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 2 & 3 & 1 & 1 & 1 \\ 1 & 2 & a & 1 & 1 \\ 2 & 3 & a+1 & b & 2 \\ 1 & 1 & 1 & 2 & c \end{pmatrix}$$

```
INd = IND(1:5)
```

```
INd = 5×1
     8
    11
    10
    15
    12
```

```
detB = simplify(det(B))
```

detB = $a\,(b-2)\,(c-1)$

```
rankB = rank(B)
```

```
rankB =
5
```

```
AMPB = [B INd]
```

AMPB =

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 8 \\ 2 & 3 & 1 & 1 & 1 & 11 \\ 1 & 2 & a & 1 & 1 & 10 \\ 2 & 3 & a+1 & b & 2 & 15 \\ 1 & 1 & 1 & 2 & c & 12 \end{pmatrix}$$

```
rankBA = rank(AMPB)
```

```
rankBA =
5
```

LA SUBMATRIZ ES UN SISTEMA COMPATIBLE DETERMINADOOOOOOOOOOO

Aplicar cramer o rref?

```
solsiscom = (B\INd)
```

solsiscom =

$$\begin{pmatrix} \frac{30a+22b+22c-21ab-20ac-14bc+13abc-32}{\sigma_1} \\ -\frac{12a+11b+11c-9ab-7ac-7bc+5abc-16}{\sigma_1} \\ -\frac{11b+11c-7bc-16}{\sigma_1} \\ -\frac{3}{b-2} \\ \frac{4b-5}{(b-2)\,(c-1)} \end{pmatrix}$$

where

$$\sigma_1 = a\,(b-2)\,(c-1)$$

no c

```
solA =solve(detB == 0, a)
```

solA = 0

```
solB = solve(detB == 0, b)
```

solB = 2

```
solC = solve(detB == 0, c)
```

solC = 1

```
%Ahora probar diferentes combinaciones
%Primero todas las condiciones
N = subs(B, [a, b, c], [solA,solB,solC])
```

N =

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 2 & 3 & 1 & 1 & 1 \\ 1 & 2 & 0 & 1 & 1 \\ 2 & 3 & 1 & 2 & 2 \\ 1 & 1 & 1 & 2 & 1 \end{pmatrix}$$

```
simplify(det(N))
```

ans = 0

```
rank(N)
```

ans =

```
4
```

```
AMPN = [N INd]
```

```
AMPN =
```

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 8 \\ 2 & 3 & 1 & 1 & 1 & 11 \\ 1 & 2 & 0 & 1 & 1 & 10 \\ 2 & 3 & 1 & 2 & 2 & 15 \\ 1 & 1 & 1 & 2 & 1 & 12 \end{pmatrix}$$

```
rankN = rank(AMPN)
```

```
rankN =
5
```

Para las tres condiciones a la vez el sistema es incompatible

```
%Probar solo con una
N = B
```

```
N =
```

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 2 & 3 & 1 & 1 & 1 \\ 1 & 2 & a & 1 & 1 \\ 2 & 3 & a+1 & b & 2 \\ 1 & 1 & 1 & 2 & c \end{pmatrix}$$

```
N = subs(N, a,sola)
```

```
N =
```

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 2 & 3 & 1 & 1 & 1 \\ 1 & 2 & 0 & 1 & 1 \\ 2 & 3 & 1 & b & 2 \\ 1 & 1 & 1 & 2 & c \end{pmatrix}$$

```
det(N)
```

```
ans =
```

$$0$$

```
rankN = rank(N)
```

```
rankN =
4
```

```
AMPN = [N INd]
```

```
AMPN =
```

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 8 \\ 2 & 3 & 1 & 1 & 1 & 11 \\ 1 & 2 & 0 & 1 & 1 & 10 \\ 2 & 3 & 1 & b & 2 & 15 \\ 1 & 1 & 1 & 2 & c & 12 \end{pmatrix}$$

```
rankAMPN = rank(AMPN)
```

```
rankAMPN =
5
```

Con a = sola el sistema es incompatible

```
N= B
```

```
N =
```

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 2 & 3 & 1 & 1 & 1 \\ 1 & 2 & a & 1 & 1 \\ 2 & 3 & a+1 & b & 2 \\ 1 & 1 & 1 & 2 & c \end{pmatrix}$$

```
N= subs(N, [a, b], [sola, solb])
```

```
N =
```

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 2 & 3 & 1 & 1 & 1 \\ 1 & 2 & 0 & 1 & 1 \\ 2 & 3 & 1 & 2 & 2 \\ 1 & 1 & 1 & 2 & c \end{pmatrix}$$

```
detN = simplify(det(N))
```

```
detN = 0
```

```
rankN2 = rank(N)
```

```
rankN2 =
4
```

```
AMPN = [N INd]
```

```
AMPN =
```

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 8 \\ 2 & 3 & 1 & 1 & 1 & 11 \\ 1 & 2 & 0 & 1 & 1 & 10 \\ 2 & 3 & 1 & 2 & 2 & 15 \\ 1 & 1 & 1 & 2 & c & 12 \end{pmatrix}$$

```
rankAMPN2 = rank(AMPN)
```

```
rankAMPN2 =
5
```

con a= sola y b solb el sistema es incompatible

```
N = B
```

```
N =
```

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 2 & 3 & 1 & 1 & 1 \\ 1 & 2 & a & 1 & 1 \\ 2 & 3 & a+1 & b & 2 \\ 1 & 1 & 1 & 2 & c \end{pmatrix}$$

```
N = subs(N, c, solc)
```

```
N =
```

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 2 & 3 & 1 & 1 & 1 \\ 1 & 2 & a & 1 & 1 \\ 2 & 3 & a+1 & b & 2 \\ 1 & 1 & 1 & 2 & 1 \end{pmatrix}$$

```
detN3 = simplify(det(N))
```

```
detN3 =
```

$$0$$

```
rankN3 = rank(N)
```

```
rankN3 =
4
```

```
AMPN = [N INd]
```

```
AMPN =
```

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 8 \\ 2 & 3 & 1 & 1 & 1 & 11 \\ 1 & 2 & a & 1 & 1 & 10 \\ 2 & 3 & a+1 & b & 2 & 15 \\ 1 & 1 & 1 & 2 & 1 & 12 \end{pmatrix}$$

```
rankAMPN3 = rank(AMPN)
```

```
rankAMPN3 =
5
```

```

```